

Requirements & Delivery Backlog

Patient management system · 31 July 2026 · repository knowledge-corner/Cmeh

This is the agreed scope of the system, broken into stories, each sized in story points and each mapped to the automated tests that prove it works. It is meant to be used two ways: to agree what is being built, and to see at a glance what is genuinely covered by tests and what is not.

Where a story has **no automated cover, it says so**. Those gaps are the most useful thing in this document — they are the places where a defect could reach the clinic unnoticed.

WHERE THE PROJECT STANDS

	STORIES	POINTS
Delivered	46	185
Partially delivered	1	5
Blocked on a clinic decision	0	0
Not started	7	34
Total scoped	54	224

215 automated tests currently pass. 7 stories carry no automated cover; each is flagged where it appears and listed again under Testing.

HOW STORY POINTS ARE USED

Points estimate relative effort and risk, not hours.

PTS	MEANING
1	Trivial — a field, a label, a one-line rule.
2	Small — one form or view, no new concepts.
3	Ordinary — a screen or a model with real behaviour.
5	Substantial — several moving parts, or rules that must not be got wrong.
8	Large — a workflow spanning roles, or logic with clinical consequences.
13	Too large — split it before starting.

- **Done** — built, and covered by tests unless a gap is noted.
- **Partial** — built but not finished; what remains is stated.
- **Blocked** — cannot proceed without a decision from the clinic.
- **Backlog** — agreed as wanted, not started.

E1 · Accounts & access control

Everyone who touches a patient record is identified, limited to their role, and recorded.

23 of 23 points delivered.

S-101 One sign-in page for every role

3 points

DONE

As any clinic user, I want a single place to sign in, so that I do not have to know which part of the system I belong to.

ACCEPTANCE CRITERIA

- Username and password, with clear errors on failure.
- Signing in while already signed in goes straight to my dashboard.
- The page states that record access is logged.

COVERED BY

- `TestLogin` (5 tests)

S-102 Land on the dashboard for my role

2 points

DONE

As a doctor, receptionist or patient, I want to arrive at my own screen after signing in, so that I never have to navigate to it.

ACCEPTANCE CRITERIA

- Doctor → today's clinic. Receptionist → queue board. Patient → appointments.
- An administrator goes to the Django admin.
- A signed-out visitor is sent to sign in.

COVERED BY

- `TestRoleRouting` (4 tests)

S-103 Roles are enforced by the server, not by hidden links

5 points

DONE

As the clinic owner, I want role limits enforced in code, so that typing a URL cannot reveal something a user should not see.

ACCEPTANCE CRITERIA

- A receptionist opening a clinical chart gets 403, not a redirect.
- A doctor cannot work the reception queue.
- A patient cannot reach any chart or billing screen.
- Restrictions hold on POST as well as GET.

COVERED BY

- `TestDoctorAccessControl` (5 tests)
- `TestEditingAccessControl` (4 tests)
- `TestReceptionQueue.test_a_doctor_cannot_work_the_queue`
- `TestBilling.test_a_patient_cannot_reach_billing`
- `TestPatientPortal.test_a_patient_cannot_reach_a_clinical_chart`

S-104 Staff accounts managed from the admin

5 points

DONE

As the clinic owner, I want to create doctor, receptionist and patient logins myself, so that I am not waiting on a developer to add staff.

ACCEPTANCE CRITERIA

- Username, password, email, contact number and role, as specified.
- Doctors carry qualification, registration number and signature for prescriptions.
- Email addresses are unique.

Test gap. No automated test — the admin screens are exercised by hand. Worth a smoke test before go-live.

S-105 Every look at a record is recorded, permanently

5 points

DONE

As the clinic owner, I want to answer “who opened this patient’s file”, so that I can respond if a patient ever asks.

ACCEPTANCE CRITERIA

- Opening a chart, creating or editing a record, printing, and signing in or out are all logged with user, time and IP.
- A failed sign-in is logged without the attempted password.
- Entries cannot be edited or deleted — including in bulk, and including from admin.
- No patient identifiers appear in application logs.

COVERED BY

- `TestAuditLogIsAppendOnly (2 tests)`
- `TestAuditLogCannotBeBulkErased (2 tests)`
- `TestEditingIsAudited (2 tests)`
- `TestPatientDashboard.test_opening_a_chart_is_written_to_the_audit_trail`
- `TestLogin.test_successful_login_is_recorded_in_the_audit_trail`
- `TestLogin.test_failed_login_is_recorded_without_the_password`
- `TestReceptionQueue.test_each_move_is_recorded_against_the_patient`

S-106 Sessions expire and cookies are secure

3 points

DONE

As the clinic owner, I want an unattended screen to lock itself, because patient records are on display in a room people walk through.

ACCEPTANCE CRITERIA

- Thirty minutes of inactivity ends the session.
- HTTPS enforced in production, with HSTS and secure, HTTP-only cookies.
- The admin is mounted at a path set by environment variable; /admin/ is a decoy.

Test gap. Configuration rather than behaviour. Verified by ``manage.py check --deploy``, which passes clean.

E2 · Patient records

One unmistakable identity per patient, and the longitudinal history that hangs off it.

16 of 16 points delivered.

S-201 Every patient gets a unique UHID

5 points

DONE

As a receptionist, I want each patient to have one permanent number, so that their whole history can be found from the number on their file.

ACCEPTANCE CRITERIA

- Format CEMH-YY-NNNNN, issued automatically, never reused, never edited.
- Two receptionists registering at the same instant cannot produce a duplicate.
- The prefix is configurable per clinic.

COVERED BY

- `TestPatientIdentifier` (4 tests)
- `TestSaving.test_uhid_is_not_editable`

S-202 Patient demographics, including guardians

3 points

DONE

As a receptionist, I want a child's guardian recorded, so that I ring the right person when confirming an appointment.

ACCEPTANCE CRITERIA

- Name, date of birth, sex, blood group, contact, address.
- Guardian name, relationship and phone for paediatric patients.
- Age shows in days, months or years as appropriate to how young they are.
- The contact number offered is the guardian's for a child, the patient's for an adult.

COVERED BY

- `TestPatientAge` (5 tests)
- `TestSaving.test_editing_patient_details_persists`

S-203 Standing background history

3 points

DONE

As a doctor, I want past history, family history and allergies on screen at every visit, so that I am not re-reading old notes to find them.

ACCEPTANCE CRITERIA

- Presenting complaints, past medical, family, birth and development, allergies, medications, surgical, lifestyle.
- One record per patient — editing never creates a duplicate.
- Allergies are shown prominently on the sidebar and on printed prescriptions.

COVERED BY

- `TestSaving.test_editing_history_updates_the_existing_record`
- `TestPatientDashboard.test_allergies_are_surfaced_on_the_sidebar`

S-204 Register a new patient at the desk

3 points

DONE

As a receptionist, I want to register a caller without leaving the booking screen, because having to go elsewhere first is what stalls a phone booking.

ACCEPTANCE CRITERIA

- Registration is reachable from the booking screen and returns to it.
- A UHID is issued on save.

COVERED BY

- `TestReceptionBooking.test_registering_a_patient_issues_a_uhid`

S-205 Find a patient by name, UHID or mobile

2 points

DONE

As clinic staff, I want to find a patient by whatever the caller gives me, so that I am not asking them to read out a number they may not have.

ACCEPTANCE CRITERIA

- Search matches UHID, first or last name, patient mobile or guardian mobile.
- The doctor can type a UHID or mobile to open a chart directly.
- UHID matching ignores case.
- An unknown value reports plainly rather than failing.

COVERED BY

- `TestReceptionBooking.test_patient_search_finds_by_uhid_and_name`
- `TestDoctorHome` (4 lookup tests)

E3 · Appointments & the clinic day

The diary reflects reality, and a patient can only move forward through the day.

31 of 31 points delivered.

S-301 A visit moves through a fixed sequence

8 points

DONE

As the clinic owner, I want the software to mirror how the clinic actually runs, so that the queue cannot get into a state that never happens in real life.

ACCEPTANCE CRITERIA

- Booked → Confirmed → Arrived → In cabin → Consulted → Billed → Completed, with Cancelled and No-show as exits.
- An out-of-order move is refused, not silently accepted.
- Arrival, cabin, consultation and completion times are stamped as they happen.
- Every change records who made it.

COVERED BY

- `TestVisitLifecycle` (9 tests)
- `TestReceptionQueue.test_an_out_of_order_move_is_refused`

S-302 Two patients can never hold one doctor at once

5 points

DONE

As a receptionist, I want double booking to be impossible, so that I never have to apologise to a patient for it.

ACCEPTANCE CRITERIA

- Enforced by the database, so it holds under simultaneous requests.
- Back-to-back appointments are fine; overlapping ones are not.
- Cancelling frees the slot again.
- Two different doctors may of course be busy at the same time.

COVERED BY

- `TestNoDoubleBooking` (4 tests)

S-303 Free slots are offered from consulting hours

5 points

DONE

As a receptionist, I want to see only times that are genuinely free, so that I can answer a caller straight away.

ACCEPTANCE CRITERIA

- Hours, slot length, working days and booking horizon are configuration.
- Booked slots disappear; cancelled ones come back.
- Closed days offer nothing and say why.
- Patients are never offered a time that has already passed; reception may be.

COVERED BY

- `TestScheduling` (5 tests)

S-304 Take a booking at the desk or over the phone

5 points

DONE

As a receptionist, I want to book a patient in a few clicks, so that I am not keeping a caller waiting.

ACCEPTANCE CRITERIA

- Search or register the patient, pick doctor and date, pick from free times.
- Every booking starts unconfirmed — the receptionist still has to ring the patient on the day, and confirming is that call.
- If someone takes the slot first, the error is readable, not a crash.
- A slot that is not on the chosen date is rejected.

COVERED BY

- `TestReceptionBooking` (6 tests)

S-305 Run the day from a queue board

8 points

DONE

As a receptionist, I want one screen showing where every patient is, so that I can call them in and send them through without a paper list.

ACCEPTANCE CRITERIA

- A column per stage, with counts.
- "To confirm" is separate from "Confirmed", so the receptionist can see exactly who she still has to telephone.
- An unconfirmed card leads with the number to ring, as a tap-to-dial link, and names the guardian to ask for when the patient is a child.
- Each card offers only the moves legal from that patient's current state.
- Waiting time counts up and is highlighted past thirty minutes.
- The board refreshes itself, because a stale board sends the wrong patient in.
- Any day can be reviewed, not just today.

COVERED BY

- `TestReceptionQueue` (5 tests)
- `TestConfirmationCallIsVisibleWork` (3 tests)

E4 · The doctor's chart

Everything known about a patient, readable at a glance and editable in place.

21 of 21 points delivered.

S-401 Open a chart by typing the UHID

3 points

DONE

As a doctor, I want to type the number from the patient's file, so that I reach their record without hunting through a list.

ACCEPTANCE CRITERIA

- UHID or mobile number both work.
- Today's queue is listed below, with whoever is in the cabin first.
- Another doctor's patients do not appear in my queue.

COVERED BY

- `TestDoctorHome` (6 tests)

S-402 A patient panel that stays on screen

3 points

DONE

As a doctor, I want identity, allergies, active problems and the latest measurements always visible, so that they are never a tab away.

ACCEPTANCE CRITERIA

- Allergies are prominent, and say so explicitly when none are recorded.
- Active problems, latest measurements, last vitals, attendance summary.
- The panel refreshes when any of it is edited, from whichever tab.

COVERED BY

- `TestPatientDashboard` (6 tests)

S-403 Five tabs, loaded without losing my place

5 points

DONE

As a doctor, I want to move between summary, notes, results, growth and prescriptions quickly, so that consultation time is not spent waiting.

ACCEPTANCE CRITERIA

- Summary, Clinical Notes, Investigations, Growth Chart, Prescriptions.
- Tabs swap without a full page reload.
- Visits are grouped into the time bands clinicians read charts in.
- Investigations are grouped by test, so a value can be followed over years.

COVERED BY

- `TestDashboardTabs` (6 tests)

S-404 Edit every element from the chart

8 points

DONE

As a doctor, I want to correct or add to any part of the record where I am reading it, so that I never go looking for a separate editing screen.

ACCEPTANCE CRITERIA

- Patient details, history, problems, investigations, notes, measurements, prescriptions and their drug lines.
- Saving refreshes the chart and panel in place.
- An invalid form redisplayes with errors rather than losing the entry.
- A record cannot be edited through another patient's URL.
- The UHID is never editable.

COVERED BY

- TestFormsOpen (4 tests)
- TestSaving (8 tests)

S-405 The chart knows which visit is in progress

2 points

DONE

As a doctor, I want the consultation I am actually in to drive the screen, so that a follow-up already in the diary does not hide it.

ACCEPTANCE CRITERIA

- The patient in the cabin takes precedence over any future booking.
- The Complete consultation action stays available.

Added after this bug was found by driving the workflow in a browser.

COVERED BY

- TestChartShowsTheVisitInProgress (2 tests)

E5 · Growth & anthropometry

Make a child's growth trajectory legible at a glance. Removable for clinics that do not need it.

37 of 37 points delivered.

S-501 Record measurements at a visit

3 points

DONE

As a paediatric endocrinologist, I want height, weight and puberty staging recorded against the date they were taken, so that trends are truthful.

ACCEPTANCE CRITERIA

- Height, weight, head circumference, waist, Tanner stage, parental heights.
- BMI is computed, never typed, so it cannot contradict height and weight.
- Age is taken at the measurement date, not today.

COVERED BY

- `TestMeasurement` (5 tests)
- `TestSaving.test_adding_a_measurement_persists_it`

S-502 Percentiles from published growth references

8 points

DONE

As a paediatric endocrinologist, I want a measurement placed on a centile, so that short stature is visible rather than inferred.

ACCEPTANCE CRITERIA

- LMS method against published reference tables.
- Verified against the SD columns published alongside those tables.
- Returns nothing rather than guessing when no reference covers the patient.
- Sex-specific; declines to chart when sex is recorded as neither.
- The standard in use is chosen by `GROWTH_REFERENCE`, and a standard whose tables are absent falls back to another — visibly, never silently.

COVERED BY

- `TestAgainstPublishedValues` (2 tests)
- `TestZScoreAndPercentile` (5 tests)
- `TestAssess` (7 tests)
- `TestStandardSelection` (8 tests)

S-503 Plot the child against the centile curves

5 points

DONE

As a paediatric endocrinologist, I want the child's line drawn over the reference family, so that crossing centiles is obvious.

ACCEPTANCE CRITERIA

- Height, weight and BMI for age; head circumference under three.
- 3rd to 97th centile curves behind the patient's own line.
- The centile is stated in words as well as drawn.

- Curves stop where the published data stops — never extrapolated.

COVERED BY

- `TestReferenceCurves` (3 tests)
- `TestDashboardTabs.test_growth_tab_plots_a_recorded_measurement`
- `TestDashboardTabs.test_growth_tab_without_measurements_shows_an_empty_state`

S-504 Mid-parental target height

2 points

DONE

As a paediatric endocrinologist, I want the target height from the parents' heights, so that I can judge whether the child is on their own track.

ACCEPTANCE CRITERIA

- Boys (father + mother + 13) / 2; girls (father + mother – 13) / 2.
- Shown only when both parental heights are known.

COVERED BY

- `TestMeasurement.test_mid_parental_height_adds_thirteen_for_a_boy`
- `TestMeasurement.test_mid_parental_height_subtracts_thirteen_for_a_girl`

S-506 Import supplied growth tables safely

3 points

DONE

As the clinic owner, I want new reference tables installed without a developer hand-editing clinical data.

ACCEPTANCE CRITERIA

- The command reads the IAP 2015 paper's own PDF, so no value is transcribed by hand; a single table may also come from a CSV.
- It refuses anything it cannot vouch for, naming the offending rows: values must rise across each row and smoothly with age.
- For BMI it also checks $P50 < 23\text{-Eq} < 27\text{-Eq}$ throughout, and that the two cut-off lines have converged on the adult 23 and 27 by eighteen years.
- The tables installed in the repository are re-validated by the test suite, so a hand edit to clinical data fails the build.

COVERED BY

- `TestASoundTableIsAccepted`, `TestTransposedColumnsAreRefused` (3 tests)
- `TestADroppedDigitIsRefused` (2 tests)
- `TestTheShapeOfTheTableIsChecked` (2 tests)
- `TestTheBmiCutoffCheck` (3 tests)
- `TestTheInstalledTablesStillPassTheirOwnChecks`

S-505 Chart against IAP 2015 Indian references

5 points

DONE

As a paediatric endocrinologist, I want Indian children charted against Indian references, so that centiles reflect the population I treat.

ACCEPTANCE CRITERIA

- WHO below five years, IAP 2015 for 5–18 — IAP's own recommendation, and the standard the clinic has chosen.
- All six tables installed from the paper: height, weight and BMI, both sexes, 5.0–18.0 years at half-year steps.
- A child of exactly 5.0 years is charted against IAP, where those charts begin; above 18 the chart falls back to CDC and says so.

- Every chart names the reference that actually produced it.

IAP Growth Chart Committee, Indian Pediatrics 2015;52:47–55, Tables II–VII.

COVERED BY

- `TestTablesAreInstalled` (2 tests)
- `TestAgainstThePublishedCentiles` (4 tests)
- `TestCurvesComeFromThePaper` (4 tests)
- `TestStandardSelection.test_the_five_year_boundary_belongs_to_iap`

S-507 Say which kind of number the reference can give

5 points

DONE

As a paediatric endocrinologist, I want to know whether a centile was computed or read off a printed curve, so that I know how much to trust the decimal places.

ACCEPTANCE CRITERIA

- WHO and CDC publish LMS parameters, so an exact centile and z-score are shown, as before.
- IAP publishes the curves only, so the child is placed in the band between two printed centiles, with an SDS interpolated between them.
- No LMS is back-fitted from the seven printed points — that would report a z-score reading as exact while carrying invisible fitting error.
- The paper's SD column is stored but never scored against: it is the sample SD, and using it would put a child on the printed 97th centile at +2.2 to +3.4 SDS instead of +1.88.
- Asking a centile reference for an exact z-score raises rather than returning an approximation.

COVERED BY

- `TestNoExactZScoreIsInvented` (4 tests)
- `TestGrowthTabShowsWhichReferenceAnswered` (5 tests)

S-508 Refuse to guess below the 3rd centile

3 points

DONE

As a paediatric endocrinologist, I want no invented number for a child off the printed scale, but I still want an SDS I can act on.

ACCEPTANCE CRITERIA

- Outside the outermost printed curve the band is still stated — 'below the 3rd centile' — but no centile or SDS is interpolated.
- A z-score from an LMS reference is supplied alongside and labelled with the source it came from, so growth-hormone decisions still have a figure.
- A child on the scale gets no companion figure, so the fallback appears only where it is needed.

WHO's 2007 5–19y reference would be a better companion than CDC for Indian children; it publishes LMS and is a drop-in when sourced.

COVERED BY

- `TestOffTheScale` (5 tests)
- `TestGrowthTabShowsWhichReferenceAnswered.test_a_child_below_the_third_centile_gets_a_labelled_companion`

S-509 Overweight and obesity from the adult-equivalent cut-offs

3 points

DONE

As a paediatric endocrinologist, I want overweight and obesity judged by the Asian cut-offs, because Indian children carry more risk at a lower BMI.

ACCEPTANCE CRITERIA

- Four bands: thinness below the 3rd centile, then normal, overweight at the 23-equivalent line and obesity at the 27-equivalent line.
- The cut-offs are age- and sex-specific — 15.7 kg/m² at five years, 23.2 at eighteen — never the adult 25 and 30.
- The 23-Eq and 27-Eq columns are excluded from centile placement, because they are cut-offs and not percentiles of anything.
- Both lines are drawn on the BMI chart, labelled, in warning colours.
- A clinic charting against WHO or CDC gets no verdict at all rather than one derived from the wrong lines.

COVERED BY

- `TestTheFourBands` (7 tests)
- `TestTheCutoffsAreNotTheAdultOnes` (4 tests)
- `TestNoVerdictWhenTheCutoffsAreUnavailable` (6 tests)
- `TestTheBmiEqColumnsAreNotCentiles` (3 tests)

E6 · Prescriptions

The doctor's instructions reach the patient on paper, and reach reception at the right moment.

13 of 13 points delivered.

S-601 Compose a prescription with medication lines

5 points

DONE

As a doctor, I want to write drug, strength, dose, frequency and duration, so that the patient leaves with unambiguous instructions.

ACCEPTANCE CRITERIA

- Multiple medication lines, plus advice, investigations advised and follow-up date.
- Existing lines can be removed.
- A new prescription is a draft — saving never issues it.

COVERED BY

- `TestPrescriptionWorkflow.test_new_prescription_starts_as_a_draft`

S-602 Issuing is a deliberate handover

3 points

DONE

As a doctor, I want to decide when a prescription is final, so that a half-written one never reaches the front desk.

ACCEPTANCE CRITERIA

- Reception can only print an issued prescription.
- The draft/issued state is visible on the chart.

COVERED BY

- `TestPrescriptionWorkflow.test_sending_to_reception_marks_it_generated`

S-603 One action ends the consultation

5 points

DONE

As a doctor, I want the fee and the prescription to go to reception together, so that the receptionist is told what to collect rather than asking.

ACCEPTANCE CRITERIA

- Records the fee, issues the prescription and moves the visit, in one step.
- Refused when the patient is not in the cabin.
- Warns when no medication has been added, but permits advice-only.
- A receptionist cannot set the fee.

COVERED BY

- `TestConsultationHandover` (4 tests)

E7 · Billing & receipts

Money is collected against what the doctor charged, and evidenced.

15 of 15 points delivered.

S-701 See what is owed before taking payment

3 points

DONE

As a receptionist, I want the charge already on screen, so that I do not have to interrupt the doctor to ask.

ACCEPTANCE CRITERIA

- Consultation fee, procedure charges and discount, with a total.
- The patient, doctor and visit date are shown alongside.

COVERED BY

- `TestBilling.test_billing_page_shows_what_is_owed`
- `TestConsultationHandover.test_a_discount_reduces_what_reception_collects`

S-702 Record payment and issue a receipt

5 points

DONE

As a receptionist, I want a numbered receipt produced automatically, so that the patient always leaves with proof of payment.

ACCEPTANCE CRITERIA

- Cash, UPI, card or bank transfer, with a reference field.
- A receipt is created on payment and records who took the money.
- Payment in full settles the visit.

COVERED BY

- `TestBilling.test_payment_in_full_issues_a_receipt_and_settles_the_visit`

S-703 Part payments stay outstanding

3 points

DONE

As a receptionist, I want a partly-paid visit to stay on my list, so that an unpaid balance is not quietly forgotten.

ACCEPTANCE CRITERIA

- The balance is shown.
- The visit is not settled until nothing is outstanding.

COVERED BY

- `TestBilling.test_a_part_payment_leaves_the_visit_unsettled`

S-704 Receipt numbers never collide

2 points

DONE

As the clinic owner, I want receipt numbers to be unique, because they are financial records.

ACCEPTANCE CRITERIA

- Format R-YY-NNNNN from a database sequence.
- Never reused, never edited.

COVERED BY

- `TestBilling.test_receipt_numbers_do_not_collide`

S-705 Check the patient out

2 points

DONE

As a receptionist, I want to close the visit when the patient leaves, so that the board reflects who is still in the building.

ACCEPTANCE CRITERIA

- Available once the visit is settled.

COVERED BY

- `TestBilling.test_checkout_completes_the_visit`

E8 · Printing

Paper that looks like it came from this clinic.

5 of 5 points delivered.

S-801 Print the prescription on letterhead

3 points

DONE

As a receptionist, I want to hand the patient a proper prescription, so that it is accepted at any pharmacy.

ACCEPTANCE CRITERIA

- Clinic name, address and telephone; doctor's name, qualification and registration number.
- Allergies printed prominently when recorded.
- A scanned signature appears when uploaded.
- Laid out for A4; printing is recorded.

COVERED BY

- `TestBilling.test_printing_a_prescription_records_it`

S-802 Print the receipt

2 points

DONE

As a receptionist, I want a printable receipt, so that the patient can claim reimbursement.

ACCEPTANCE CRITERIA

- Receipt number, date, itemised charges, amount received, method.
- Balance shown if any remains.

COVERED BY

- `TestBilling.test_printing_a_receipt_shows_the_number`

E9 · The public page

One page that tells people what the clinic does and gets them to telephone or send a WhatsApp message. Patients never sign in.

8 of 8 points delivered.

S-901 A single public page for the clinic

5 points

DONE

As somebody looking for an endocrinologist, I want to see what this clinic treats and who its doctors are, so that I know it is the right place to ring.

ACCEPTANCE CRITERIA

- Clinic name, address, consulting hours and both consultants with their qualifications.
- What the clinic treats, in adult and paediatric columns, matching the printed brochure.
- Readable on a phone, which is how most people will find it.
- The only page in the system that search engines may index; everything behind the login stays hidden from them.
- No patient information appears anywhere on it.

COVERED BY

- `TestPublicPage` (9 tests)
- `TestPublicPageLeaksNothing` (2 tests)
- `TestEverythingElseStillNeedsALogin` (2 tests)

S-902 Call and WhatsApp buttons that actually book

3 points

DONE

As somebody wanting an appointment, I want to reach the clinic in one tap, so that I do not have to fill in a form and wait.

ACCEPTANCE CRITERIA

- Tap-to-dial and WhatsApp buttons, at the top of the page and again at the end.
- The WhatsApp link carries the country code and a message already typed.
- The page says plainly that bookings are by telephone, not online.
- Which number receives them is configurable per clinic.

COVERED BY

- `TestPublicPage.test_offers_a_telephone_link`
- `TestPublicPage.test_offers_a_whatsapp_link_in_international_format`

S-903 Patient portal — logins, appointment list, online booking

0 points

WITHDRAWN

As a patient, I want to see my appointments and book online.

ACCEPTANCE CRITERIA

- Built and working, then removed at the clinic's request.
- Reception takes every booking; patients call or send a WhatsApp message.

Withdrawn, not forgotten. Recorded here so the decision is visible: the clinic decided patients should not book online, so a working portal was deleted rather than left behind an unused login.

COVERED BY

- TestNoPatientPortal (2 tests) – asserts the routes are gone

E10 · Platform, reuse & deployment

Runs reliably for this clinic, and can be re-skinned for the next one without a fork.

16 of 21 points delivered.

S-1001 12-factor project on PostgreSQL, in Docker

5 points

DONE

As a developer, I want configuration in the environment, so that the same image runs any clinic.

ACCEPTANCE CRITERIA

- Separate dev, production and test settings.
- Database, secrets, hosts and branding from environment variables.
- docker compose up gives a working stack from nothing.
- Production refuses to start with a default secret key or empty hosts.

Test gap. No automated test. Verified by rebuilding the database from empty and running `check --deploy`, which passes.

S-1002 Realistic demo data on demand

3 points

DONE

As a developer or reviewer, I want lifelike data in one command, so that screens can be judged against something resembling a real practice.

ACCEPTANCE CRITERIA

- Five patients with years of visits, results and measurements.
- Re-runnable; refuses to run with DEBUG off unless forced.
- Identifiers restart so demo output is reproducible.

Test gap. No automated test — exercised manually on every rebuild.

S-1003 Re-skin for another clinic without forking

3 points

DONE

As the clinic owner, I want a second clinic to differ only in branding and features, so that core fixes still merge down.

ACCEPTANCE CRITERIA

- Colours in one CSS variables file.
- Name, address, UHID prefix and hours in one settings file.
- Logo and letterhead in one templates directory.

Test gap. No automated test. Verified by editing theme.css and confirming the whole interface re-skins.

S-1004 Speciality features are removable modules

5 points

DONE

As the clinic owner, I want to switch off features another clinic does not need, so that an orthopaedic practice never sees growth charts.

ACCEPTANCE CRITERIA

- Removing 'growth' from OPTIONAL_APPS removes its models, admin and tab together.
- No core app imports from a speciality app.
- The application still boots with it removed.

Test gap. No automated test. Verified by hand. Worth automating — this is the property the whole multi-clinic strategy rests on.

S-1005 Deploy to the clinic's own server

5 points

PARTIAL

As the clinic owner, I want the system running on our own domain, so that staff can start using it.

ACCEPTANCE CRITERIA

- DigitalOcean droplet in Bangalore with managed PostgreSQL.
- Caddy for automatic HTTPS.
- Automated backups with point-in-time recovery, and a tested restore.
- Domain pointed and certificate issued.

Test gap. Compose files, Caddyfile and production settings are written and checked. Nothing has been deployed — no server, domain or database exists yet.

Backlog — agreed but not started

7 stories · 34 points

S-1102 Appointment reminders by SMS or WhatsApp

8 points

BACKLOG

As a receptionist, I want patients reminded automatically, so that fewer fail to attend.

Needs a provider decision and a background job runner — neither exists yet.

S-1103 Clinical form builder

8 points

BACKLOG

As a doctor, I want to add my own questions to the consultation form without a developer.

The FormDefinition model and the JSON field it drives are built; there is no screen to edit definitions, so new fields still need a developer.

S-1104 Daily collection and attendance report

5 points

BACKLOG

As the clinic owner, I want a daily summary of patients seen and money taken.

All the underlying data is recorded; nothing reports on it yet.

S-1105 Doctor availability and leave

5 points

BACKLOG

As a receptionist, I want to block out leave and vary consulting hours per doctor.

Hours are currently one setting for the whole clinic.

S-1106 Upload investigation reports from the chart

3 points

BACKLOG

As a doctor, I want to attach the lab's PDF to a result.

The file field exists on the model and works in admin, but is not on the chart's add-result form.

S-1108 Lock accounts after repeated failed sign-ins

2 points

BACKLOG

As the clinic owner, I want brute-force attempts blocked.

Failed attempts are already logged. Recommended before go-live.

S-1109 Backup and restore runbook

3 points

BACKLOG

As the clinic owner, I want a tested procedure for restoring patient data.

Managed PostgreSQL provides the backups; the restore has never been rehearsed. This is the highest-risk gap once real patients are in the system.

Testing

Automated. The suite runs in seconds against a real PostgreSQL database — `pytest` from the project root. The figure above is counted from the suite itself when this document is generated, not written down.

Browser-driven. The full booking-to-receipt path, and the public page, have been driven through a real browser. This found four bugs the unit tests had not — including one where a booked follow-up hid the doctor's Complete consultation action.

Not yet covered. Django admin screens, the deployment configuration, the removable-module guarantee, and the demo-data command. Each is noted against its story.

Clinical validation. The growth-chart maths is checked against published reference tables, but no clinician has yet reviewed the system against real cases.

STORIES WITH NO AUTOMATED COVER

These are the places a regression would not be caught.

STORY	WHAT IS MISSING
S-104 Staff accounts managed from the admin	No automated test — the admin screens are exercised by hand. Worth a smoke test before go-live.
S-106 Sessions expire and cookies are secure	Configuration rather than behaviour. Verified by `manage.py check --deploy`, which passes clean.
S-1001 12-factor project on PostgreSQL, in Docker	No automated test. Verified by rebuilding the database from empty and running `check --deploy`, which passes.
S-1002 Realistic demo data on demand	No automated test — exercised manually on every rebuild.
S-1003 Re-skin for another clinic without forking	No automated test. Verified by editing theme.css and confirming the whole interface re-skins.
S-1004 Speciality features are removable modules	No automated test. Verified by hand. Worth automating — this is the property the whole multi-clinic strategy rests on.
S-1005 Deploy to the clinic's own server	Compose files, Caddyfile and production settings are written and checked. Nothing has been deployed — no server, domain or database exists yet.

RUNNING THE TESTS

```
pytest # all 215
pytest tests/test_workflow.py # the clinic day, booking to receipt
pytest tests/test_growth_reference.py # percentile maths vs published tables
```

Open decisions for the clinic

1. Where is it hosted, and when do we go live?

Recommended: DigitalOcean Bangalore with managed PostgreSQL, for India data residency and automated backups (S-1005).

2. What else does the receptionist capture at check-in?

The mechanism for clinic-specific fields exists, but no fields have been agreed (S-1103).

3. Do patients get portal logins, and who issues them?

Today they are created one at a time in the admin (S-1107).

4. Should WHO's 2007 5–19 year reference be added alongside IAP?

It publishes LMS, so it would give a continuous SDS for a child below the 3rd centile, where the IAP tables stop. CDC fills that role today and is labelled as doing so (S-508). A decision for Dr. Vrushali.